Birzeit University
Faculty of Engineering and Technology
Department of Electrical and Computer Engineering
ENCS3320 – Computer Networks (Term 1252)
Project2 (*Network Design*) – Due June 19, 2026

A) Overview

This project is designed to provide students with practical experience in computer network design, IP addressing, subnetting, device configuration, and static routing using Cisco Packet Tracer. The project must be implemented using **Cisco Packet Tracer (any version)**. Students are allowed to work in groups of **a maximum of three students** from any section.

B) Submission Instructions

Each group must submit:

- One Packet Tracer file only (**.pkt**)
- The file must be named using the students' IDs in the following format: *ID1_ID2_ID3.pkt*
  Example: 1210345_1210678_1210890.pkt
  Groups with fewer than three students should include only the participating student IDs in the filename.

C) Project Objectives:

By completing this project, students will be able to:

- Design and implement a small computer network using Cisco Packet Tracer.
- Apply subnetting concepts in real network scenarios.
- Configure routers, switches, servers, and end devices.
- Configure static IP addressing and DHCP services.
- Implement static routing between multiple routers.
  - Configure and test common network services such as: DHCP, DNS, HTTP (Web Server), and Email Service.
- Verify network connectivity and troubleshoot networking issues.
- Improve teamwork and technical documentation skills.

D) Network Topology Requirements:

Students must create and implement the exact topology shown in **Figure 1**. The required topology contains the following devices:

- 3 Routers (Router-PT)
- 2 Switches (Switch-PT)
- 3 Servers (Server-PT)
- 4 End Devices (PCs (PC-PT) and/or Laptops (Laptop-PT))

The topology must be fully connected and configured according to the project requirements. Students are expected to correctly connect all devices, configure the necessary interfaces, and ensure proper communication between all networks.
You may organize the placement of devices for better readability; however, the logical topology and required devices shown in **Figure 1** must be maintained.

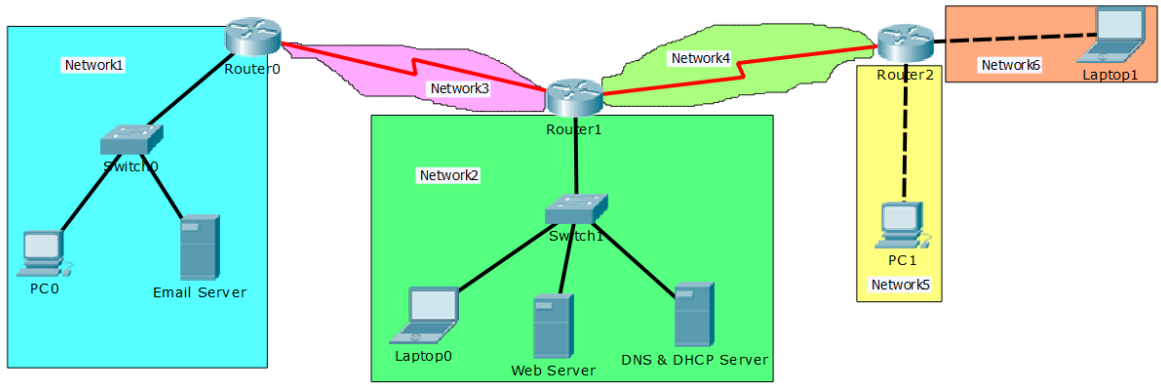


Figure 1: Network Topology.

E) Tasks:

Part 1 – IP Addressing and Subnetting

Assume that the main network address is:
192.170.X.0
Where: X represents the third and fourth digits of the student ID of any member in the group.
Example:

If the chosen student ID is: 1213456. Then: **X** = 34 Therefore, the network becomes: **192.170.34.0/24**

Requirements:
You are required to:

1. Design a suitable subnetting scheme for the entire network based on the given topology in **Figure 1** and the numbers given in **Table 1**.
2. Divide the network into appropriate subnets based on the given topology, number of requested IPs, and the predefined IP from your ID.
3. Assign valid IP addresses to (Routers, PCs/Laptops, and Servers) based on the results network IPs from the subnetting part.
4. Clearly document all subnet masks, default gateways, and assigned IP addresses.

Table 1: Number of IPs for each network in the given topology.

Networks	IPs (including the devices in Figure 1)
Network1	**32**
Network2	**63**
Network3	**4**
Network4	**4**
Network5	**14**
Network6	**16**

Part 2 – Device Configuration

Configure all network devices properly, including:

Routers:

- Interface IP addresses
- Interface activation (no shutdown)

PCs/Laptops:

- Static or dynamic IP configuration as required

Servers: configure the following services

a) DHCP Server:

- Create DHCP pool with name (Pool_stdID) for the required network
- Configure:
  - Default gateway
  - DNS server
  - IP range (excluding the first ten IPs)

b) DNS Server:

- Configure DNS resource records (A type-RRs) for the **web** and **mail** services

c) HTTP Server:

- Enable the web service with domain name:
  - *www.FamilyNameFirstName.com*, for example: *www.nemeribrahim.com*
- Create a simple webpage version from the one that you created in the first project.

d) Email Server:

- Configure:
  - Domain name: *mail.FamilyNameFirstName.com* (i.e. *mail.nemeribrahim.com*)
  - User accounts (two users as in **Table 2**)

Table 2: Email configuration for the users.

#	PC/Laptop	Name	Password
1	PC0	Last name of student1	Most significant 4 digits of student1
2	Laptop1	First name of student1	Least significant 4 digits of student1

Part 3 – Static Routing

Configure static routing between all routers to ensure full connectivity between all networks. All devices across different networks must be able to communicate successfully. For more details on how to do static routing you can check the files under the resources in section (I).

Part 4 – Testing and Verification

You must verify and demonstrate the following:

- Successful **ping** connectivity between all networks
- Proper DHCP address assignment
- Correct DNS name resolution
- Access to the web server using a browser
- Successful sending and receiving of emails
- Check the paths between all end devices using the **tracert** command
- Full communication between all PCs/Laptops and servers

F) Important Notes

- The topology must be cleaned, organized, and clearly labeled.
- All routers, switches, PCs/Laptops, servers, interfaces, IP addresses, and subnet information must be labeled inside the Packet Tracer file.
- Add notes/comments inside the .pkt file explaining:
  - Subnetting details
  - Network design
  - IP allocation
  - Routing configuration
- Use meaningful device names and hostnames.
- Any missing labels or unclear configurations may result in grade deductions.
- The submitted .pkt file must run correctly without errors.
- Incorrect file naming may result in mark deductions.
- Late submissions will not be accepted.

G) Submission Notes

- Submit the .pkt file only through ITC **before end of June 19, 2026**.
- Make sure the file opens and works correctly before submission.
- Only one submission per group is required. If multiple submissions are received from the same group, only the last submission will be considered.
- Please make sure to follow all file naming and submission rules exactly as specified before uploading your file. There will be marks allocated for following the required submission instructions.
- Every group member is responsible for understanding all parts of the project.

H) Grading

This project is worth **7%** of the total grade.

I) Resources

Drive link:
https://drive.google.com/drive/folders/1DW6GgM6zO2qmGhnwObUVfEVBqlOziRkw?usp=drive_link